Physics - Graduate Level Assessment 24 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. The atomic mass of the 4 He atom is 4.002603 u. Find the binding energy of the 4 He nucleus in MeV.
   (a) 33.9
   (b) 29.8
   (c) 27.5
   (d) 31.2
   (e) 24.3
2. Calculate the self-inductance of a coil if a rate of change of current of 5 amperes per second produces a backemfof 2 volts.
   (a) 0.4 henry
   (b) 0.3 henry
   (c) 0.2 henry
   (d) 0.6 henry
   (e) 0.7 henry
3. What happens to the force of gravitational attraction between two small objects if the mass of each object is doubled and the distance between their centers is doubled?
   (a) It is increased by two times.
   (b) It remains the same.
   (c) It becomes zero.
   (d) It is reduced to half.
   (e) It is increased by eight times.
4. What is the focal length, of a combination of two thin convex lenses, each of focal length f, placed apart at a distance of (2 f / 3)?
   (a) (f / 3)
   (b) (f / 2)
   (c) (5 f / 3)
   (d) (6 f / 5)
   (e) (3 f / 4)
5. Why are the season's in Mars' southern hemisphere so extreme?

(a) because Mars' axis is less tilted than the Earth's

(b) because Mars has a smaller size than the Earth

(c) because Mars has a slower rotation speed than the Earth

(d) because Mars has more carbon dioxide in its atmosphere than the Earth

(e) because Mars has a shorter year than the Earth

True / False (10 marks)

Answer True or False

6. True or False: The ratio of forward-bias to reverse-bias currents at 1.5 V is -6 e+25 under room temperature conditions.

7. True or False: The angular size of an object can be used to estimate its distance from an observer, indicating the rock was about 6 meters away from the rover.

8. True or False: The focal length of the lens is -20 inches, indicating it is a diverging lens that produces virtual images.

9. True or False: The transmittance of a thin film with half the optical density of a coating that absorbs 88% of light is 34.6%.

10. True or False: The best approximation for the surface temperature of the Sun is approximately 6000 K.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the change in entropy when 100 g of water at $0^{i} smixedwith 50 gofwaterat 50^{\cdot} Includecalculatio$

11. Calculate the electrostatic force acting on a 10.0 g block with a charge of $+8.00 \times 10$ C placed in the electric field $\vec{E} = (3000\hat{\imath} - 600\hat{\jmath})$N/C. Discuss the steps in your calculation and the physical principles involved.

End of Paper